

◆ WEEK 2 — SESSION 6

DATABASES

[Amazon RDS vs DynamoDB](#)

PREMIUM READ & TEACH NOTES

AWS CLOUD PRACTITIONER BOOTCAMP

◆ SLIDE 1 — PREMIUM COVER SLIDE

DATABASE ARCHITECTURE FOUNDATIONS

[Amazon RDS vs DynamoDB](#)

 [INSTRUCTOR DELIVERY SCRIPT](#)

“Welcome everyone to today’s session on Database Architecture Foundations.

Today we are entering one of the most important areas of cloud computing — databases.

Almost every modern application depends on databases.

Banking systems.

Hospital systems.

E-commerce platforms.

Mobile applications.

Social media systems.

Without databases, modern technology cannot function properly.

Today’s goal is not simply memorizing AWS services.

Our goal is understanding:

how businesses store data,

why different database models exist,

and how cloud architects decide which database to use.

This is extremely important because poor database decisions create:

- poor performance,
- high costs,
- scaling problems,
- and operational failures.

By the end of this session, you should clearly understand:

- what databases are,
 - the difference between relational and NoSQL databases,
 - how Amazon RDS works,
 - how DynamoDB works,
 - and why architecture decisions matter.”
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◆ SLIDE 2 — WHAT IS A DATABASE?

What Is a Database?

TEACHING NOTES

“A database is simply a system used to store and organize information.

Think about your phone contacts.

Your phone stores:

- names,
- phone numbers,
- emails,
- and addresses.

That is a simple database.

Now imagine businesses operating with millions of customers.

They must store:

- customer records,
- transactions,
- passwords,
- payments,
- inventory,
- medical records,
- and analytics.

Databases help businesses organize and retrieve this information quickly.

Without databases:

- ✗ applications cannot remember users
- ✗ websites cannot store information
- ✗ systems cannot process transactions

Databases are the memory systems of modern applications.”

SIMPLE ANALOGY

“Think of a database like a massive digital library.

If books are scattered randomly everywhere, finding information becomes difficult.

But if books are organized properly:

- ✓ information becomes easy to locate
- ✓ systems become efficient
- ✓ operations become reliable

Databases work the same way.”

◆ SLIDE 3 — WHY DATABASES MATTER

Why Databases Matter

TEACHING NOTES

“Modern businesses depend on data.

Data drives:

- decisions,
- customer management,
- operations,
- automation,
- and analytics.

For example:

A banking app must instantly retrieve your balance.

An e-commerce platform must instantly retrieve product information.

A hospital system must instantly retrieve patient records.

Databases make this possible.

As businesses grow, databases must also scale.

This is why cloud databases became extremely important.

AWS allows organizations to build scalable database systems without purchasing physical infrastructure.”

REAL-WORLD EXAMPLE

“Imagine Takealot during Black Friday.

Millions of customers are:

- browsing products,
- adding items to carts,
- making payments.

Behind the scenes, databases are constantly working.

Without scalable databases, the platform would crash.”

◆ SLIDE 4 — RELATIONAL DATABASES

Relational Databases

TEACHING NOTES

“Relational databases organize data into tables.

Tables contain:

- rows
- columns

Very similar to Excel spreadsheets.

For example:

Customer ID Name Email

Relational databases are highly structured.

Relationships exist between different pieces of information.

For example:

A customer account may connect to:

- transactions,
- orders,
- payments,
- and invoices.

Relational databases are excellent when:

- ✓ accuracy matters
- ✓ consistency matters
- ✓ relationships matter

This is why banks heavily use relational databases.”

POWERFUL ANALOGY

“Think of relational databases like a highly organized filing cabinet inside a government office.

Every document:

- ✓ has a correct folder
- ✓ follows strict structure
- ✓ connects to other records

Nothing is random.

This structure improves reliability and governance.”

◆ SLIDE 5 — INTRODUCTION TO AMAZON RDS

Amazon RDS

TEACHING NOTES

“Amazon RDS stands for:

Relational Database Service.

It is AWS’s managed relational database platform.

Before cloud computing, companies had to:

- buy database servers,
- install operating systems,
- patch software,
- configure backups,
- maintain hardware.

This was expensive and difficult.

Amazon RDS simplifies this process.

AWS manages much of the infrastructure automatically.”

◆ WHAT AWS MANAGES

“AWS helps manage:

- ✓ backups
- ✓ patching
- ✓ availability
- ✓ monitoring
- ✓ maintenance

This allows engineers to focus more on applications and business operations.”

SIMPLE ANALOGY

“Think of Amazon RDS like renting a professionally managed office building.

You focus on your business operations.

The building management handles:

- electricity,
 - maintenance,
 - security,
 - repairs,
 - and infrastructure.”
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◆ SLIDE 6 — RDS DATABASE ENGINES

RDS Database Engines

TEACHING NOTES

“Amazon RDS supports multiple database engines.

Examples include:

- MySQL
- PostgreSQL
- MariaDB
- Oracle
- Microsoft SQL Server

Different businesses prefer different database engines based on:

- existing systems,
 - licensing,
 - performance needs,
 - and technical requirements.”
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IMPORTANT UNDERSTANDING

“AWS is not forcing businesses to completely rebuild their systems.

RDS allows organizations to move existing database technologies into the cloud more easily.”

◆ SLIDE 7 — WHAT IS NOSQL?

Introduction to NoSQL

TEACHING NOTES

“NoSQL databases are designed differently.

Instead of strict tables and relationships, NoSQL databases focus on:

- ✓ flexibility
- ✓ speed
- ✓ scalability

NoSQL systems are designed for modern cloud-native workloads.

They work extremely well for:

- mobile apps,
- gaming systems,
- IoT platforms,
- and real-time applications.”

SIMPLE ANALOGY

“Imagine relational databases as a perfectly organized office filing system.

Now imagine NoSQL as a massive smart warehouse that automatically adapts based on demand.

It may look less structured,
but it scales much faster.”

◆ SLIDE 8 — INTRODUCTION TO DYNAMODB

Amazon DynamoDB

TEACHING NOTES

“Amazon DynamoDB is AWS’s fully managed NoSQL database service.

DynamoDB is built for:

- ✓ ultra-fast performance
- ✓ massive scalability
- ✓ serverless architecture
- ✓ cloud-native systems

One major advantage is automatic scaling.

If millions of users suddenly connect,
DynamoDB can scale rapidly.”

REAL-WORLD EXAMPLE

“Imagine a global mobile game.

Millions of players are updating scores every second.

A traditional relational system may struggle.

DynamoDB is designed for this kind of massive-scale workload.”

◆ SLIDE 9 — WHAT SERVERLESS REALLY MEANS

Serverless Databases

TEACHING NOTES

“One of the most misunderstood cloud terms is:

‘Serverless.’

Serverless does NOT mean there are no servers.

Servers still exist.

It means AWS manages the infrastructure for you.

You focus on:

- ✓ applications
- ✓ logic
- ✓ development

AWS handles:

- ✓ scaling
- ✓ maintenance
- ✓ availability
- ✓ infrastructure operations”

SIMPLE ANALOGY

“Think of serverless like using Uber.

You still use a car.

But you are not responsible for:

- maintenance,
- fuel,
- repairs,
- or parking.

The platform manages the operational complexity.”

◆ SLIDE 10 — RDS VS DYNAMODB

RDS vs DynamoDB

TEACHING NOTES

“This is where architecture thinking becomes important.

Neither service is ‘better.’

The correct question is:

Which service matches the workload?

Amazon RDS is excellent for:

- ✓ structured systems
- ✓ financial records
- ✓ transactional consistency

DynamoDB is excellent for:

- ✓ high-scale workloads
- ✓ real-time systems
- ✓ flexible cloud-native applications”

POWERFUL ANALOGY

“Choosing a database is like choosing transportation.

A sports car is excellent for speed.

A truck is excellent for heavy loads.

A bus is excellent for transporting many people.

None are universally better.

The best choice depends on the workload.”

◆ SLIDE 11 — WHEN TO USE RDS

When To Use Amazon RDS

TEACHING NOTES

“Use Amazon RDS when systems require:

- ✓ strong relationships
- ✓ structured records
- ✓ transactional consistency
- ✓ reliability

Examples include:

- banking systems
- payroll systems
- ERP systems
- healthcare records

These systems cannot tolerate inconsistent data.”

IMPORTANT UNDERSTANDING

“If money transfers fail incorrectly inside a banking system, the consequences become extremely serious.

This is why relational consistency matters.”

◆ SLIDE 12 — WHEN TO USE DYNAMODB

When To Use DynamoDB

TEACHING NOTES

“Use DynamoDB when applications require:

- ✓ rapid scaling
- ✓ flexible structures
- ✓ real-time performance
- ✓ event-driven operations

Examples:

- gaming systems
 - social platforms
 - IoT systems
 - mobile applications”
-

IMPORTANT ARCHITECTURE THINKING

“Modern cloud architecture is about selecting the correct tool for the correct problem.

Strong architects understand workload behavior before selecting technology.”

◆ SLIDE 13 — SECURITY & GOVERNANCE

Database Security & Governance

TEACHING NOTES

“Databases contain highly sensitive information.

Examples:

- financial data
- customer records
- passwords
- patient records

This means security becomes extremely important.

Organizations must think about:

- ✓ encryption
- ✓ backups
- ✓ monitoring
- ✓ IAM permissions
- ✓ disaster recovery”

REALITY CHECK

“If attackers compromise a database, entire businesses can be damaged.

This is why cloud governance matters.”

◆ SLIDE 14 — ARCHITECTURE THINKING

Architecture Decision Thinking

TEACHING NOTES

“Cloud architects do not randomly choose services.

They evaluate:

- ✓ scalability
- ✓ cost
- ✓ performance
- ✓ operational complexity
- ✓ business requirements

A wrong database decision may create:

- ✗ slow applications
 - ✗ high operational costs
 - ✗ poor scalability
 - ✗ system instability”
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FINAL ANALOGY

“Architecture is like constructing a building.

If the foundation is weak,
the entire structure becomes unstable.

Database selection is part of that foundation.”

◆ SLIDE 15 — PREMIUM CLOSING SLIDE

DATABASE ARCHITECTURE FOUNDATIONS

Amazon RDS vs DynamoDB

CLOSING DELIVERY SCRIPT

“Today we explored one of the most important areas in cloud computing — databases.

You learned:

- ✓ what databases are
- ✓ why businesses depend on them
- ✓ the difference between relational and NoSQL systems
- ✓ how Amazon RDS works
- ✓ how DynamoDB works
- ✓ why architecture decisions matter

Most importantly:

You began thinking like cloud architects instead of simply cloud users.

Remember:

Strong cloud practitioners know services.

Future architects understand:

WHY services exist,

WHEN to use them,

and HOW they affect business systems.

Excellent work today.”